

MONTREAL MCTAVISH, QC, Canada

WMO: 716120

Lat: 45.5050N

Lon: 73.5792W

Elev: 238

StdP: 14.57

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.9	-2.3	-18.9	2.0	-5.4	-14.1	2.6	-0.9	12.2	15.0	11.5	15.4	4.8	270	0.430

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.4	86.4	71.2	83.6	69.5	81.2	68.1	73.9	82.7	72.1	80.0	70.5	78.0	4.6	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.0	115.1	79.1	69.3	108.8	77.3	67.7	102.7	75.6	37.7	82.8	36.0	80.1	34.6	78.0	82.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.9	9.6	8.6	DB	-12.0	91.0	5.0	2.4	-15.6	92.8	-18.5	94.2	-21.3	95.6	-24.9	97.3	(4)
(5)			WB	-12.7	77.4	4.7	2.0	-16.2	78.8	-18.9	80.0	-21.6	81.1	-25.0	82.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	46.3	17.9	20.6	30.4	44.2	57.8	66.8	71.7	70.1	62.6	49.8	37.4	25.1		
	DBStd	20.79	12.02	10.66	10.27	8.58	7.91	6.63	5.02	5.21	7.17	7.92	8.74	10.50		
	HDD50	3927	996	824	611	214	18	0	0	0	2	104	385	773		
	HDD65	7400	1461	1244	1073	625	250	56	6	14	129	475	829	1238		
	CDD50	2585	0	0	3	39	260	505	672	623	379	97	7	0		
	CDD65	583	0	0	0	1	27	111	212	172	57	3	0	0		
	CDH74	3494	0	0	1	15	190	713	1327	938	302	8	0	0		
	CDH80	785	0	0	0	2	37	172	316	195	63	0	0	0		
(14) Wind	WSAvg	4.2	4.5	4.5	4.7	4.8	4.5	4.0	3.7	3.7	3.9	4.0	4.1	4.3		
(15) Precipitation	PrecAvg	43.1	2.9	2.9	3.2	3.1	3.2	3.7	4.0	4.1	3.7	3.3	4.0	4.3		
	PrecMax	54.3	5.6	7.1	5.4	6.5	6.5	5.9	7.6	8.1	7.5	5.7	9.2	8.0		
	PrecMin	35.0	0.7	0.9	0.8	0.9	0.6	0.8	1.6	1.2	0.9	0.8	1.6	1.9		
	PrecStd	5.3	1.2	1.6	1.3	1.1	1.5	1.4	1.5	1.8	1.6	1.3	1.5	1.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.7	46.8	62.9	75.6	84.5	88.8	90.1	88.4	86.5	74.6	62.2	51.6		
		MCWB	47.1	42.2	54.0	60.4	66.5	72.7	74.3	72.7	71.4	63.3	56.7	48.9		
	2%	DB	42.6	42.1	52.5	67.2	79.2	84.6	86.3	84.5	81.1	69.3	57.2	45.8		
		MCWB	40.7	38.4	44.5	52.9	62.9	69.7	71.5	70.2	68.6	60.3	52.7	43.1		
	5%	DB	38.4	38.7	47.1	61.9	74.9	81.4	83.5	81.9	77.2	64.8	53.2	41.7		
		MCWB	36.1	35.5	40.8	50.0	60.1	67.7	70.1	68.9	66.9	57.1	48.7	39.2		
	10%	DB	34.6	35.7	43.5	57.1	70.7	78.0	80.9	79.3	73.8	61.1	49.5	38.3		
		MCWB	32.4	33.1	38.1	46.7	58.5	65.2	68.4	67.6	65.2	54.7	45.0	36.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.3	43.4	55.4	62.0	69.5	74.7	77.0	75.5	73.5	66.2	58.0	49.3		
		MCDB	48.5	46.2	62.0	72.8	80.0	85.8	87.4	84.8	82.3	72.3	60.4	51.4		
	2%	WB	40.7	38.9	45.7	55.9	66.2	72.2	74.5	72.9	70.7	61.6	53.3	43.3		
		MCDB	42.4	41.5	50.7	63.5	74.8	81.9	83.0	81.0	78.0	67.4	56.9	45.5		
	5%	WB	36.2	35.9	41.7	51.4	63.3	69.9	72.4	71.1	68.4	58.3	49.4	39.5		
		MCDB	38.4	38.2	46.3	59.5	71.4	77.7	80.0	78.2	75.6	63.7	52.4	41.3		
	10%	WB	32.7	33.3	38.7	48.1	60.1	67.8	70.6	69.6	66.2	55.5	45.4	36.2		
		MCDB	34.5	35.5	42.8	56.3	67.8	75.5	78.2	76.5	72.9	60.2	48.9	38.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	12.5	12.8	12.8	14.6	15.9	15.2	14.4	14.4	14.5	12.0	10.4	10.5		
		MCDBR	14.9	14.2	16.8	22.1	20.6	18.8	16.6	16.5	17.3	17.0	15.7	12.9		
		MCWBR	15.0	12.9	12.2	13.8	11.0	10.0	8.7	8.1	9.3	10.8	13.6	12.4		
	5% WB	MCDBR	15.0	13.9	16.0	20.2	17.8	16.3	14.8	14.9	15.8	15.5	15.4	12.8		
		MCWBR	15.2	13.3	13.1	14.3	11.8	10.1	9.2	8.7	9.8	11.3	14.3	12.8		
(40) Clear-Sky Solar Irradiance	taub		0.258	0.288	0.330	0.362	0.388	0.439	0.428	0.413	0.373	0.335	0.306	0.292		
	taud		2.348	2.257	2.292	2.360	2.345	2.207	2.245	2.282	2.383	2.494	2.489	2.306		
	Ebn at Noon		273	283	285	285	279	264	266	265	267	265	253	242		
	Edh at Noon		23	32	36	36	38	44	42	39	33	25	21	23		
(44) All-Sky Solar Radiation	RadAvg		458	760	1154	1385	1624	1787	1796	1589	1258	726	439	318		
	RadStd		51	79	94	161	153	135	134	74	103	64	56	27		

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+166	N/A
(47)	Regional (43 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page